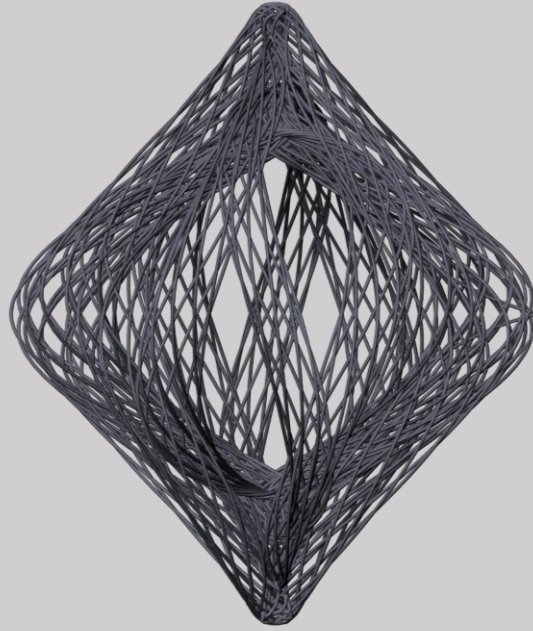




portefølje

N I M A Z A H I R I



PERSONALIA: NIMA ZAHIRI

Byggingeniør, ph.d. i trekonstruksjon

E-post: zahiri.nima@gmail.com

Telefon: +47 459 15 627

LinkedIn: [linkedin.com/in/nimazahiri/](https://www.linkedin.com/in/nimazahiri/)

Nettside: nimazahiri.github.io

Bernau bei Berlin, Tyskland

NØKKEL- KVALIFIKASJONER

- Flere års internasjonal erfaring
- Akademisk utdanning på høyt nivå
- Teknisk faglig portefølje
- Bred programvarekunnskap
- Solid produksjonskompetanse
- Sterke kommunikasjonsevner

AKADEMISK UTDANNING PÅ HØYT NIVÅ

Doktorgrad i ingeniørvitenskap

2023 byggteknikk av
2020 trekonstruksjoner
NTNU i Gjøvik, NO

Master i ITECH

2019 Integrative Technologies and
2017 Architectural Design Research
University of Stuttgart, DE

Bachelor i ingeniørfag

2011 Konstruksjons-
2006 og byggteknikk
Razi University, IR

BRED RELEVANT PROGRAMVAREKUNNSKAP

Parametrisk design – Grasshopper
Programmering – Python, C#
BIM-modellering – Revit
FEM-analyse – Karamba3D , Fusion 360
Trekonstruksjon – cadwork, SEMA
Trebearbeiding – Hundegger-Cambium

FLERE ÅRS INTERNASJONAL ARBEIDSERFARING

Digitaliseringsingeniør (trebygg)

d.d. Prosjektering av modulære
2023 trekonstruksjoner
timpla GmbH, Eberswalde, DE

Computational Designer / 3D-spesialist

2020 Computational design av
komplekse betongarmeringer
Werner Sobek AG, Stuttgart, DE

Vitenskapelig assistent

2019 Forskningsstøtte i
2017 doktorgradsprosjekter
University of Stuttgart, DE

Prosjektingeniør

2016 Bygningsinformasjonsmodellering
2014 og tegning av konstruksjoner
T.AHA Architecture, Teheran, IR

SPRÅKFERDIGHETER – FIRE SPRÅK

Persisk – morsmål
Engelsk – flytende på akademisk nivå
Tysk – avansert i arbeidssammenheng
Norsk – selvstendig i dagligtale

Denne porteføljen handler om computational design,
integreerte teknologier og digital fabrikasjon.

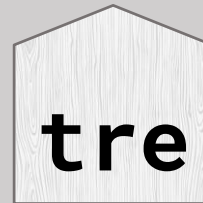
Den består av to kapitler:

KAPITTEL I



teknologiske
prosjekter

KAPITTEL II



trebaserte
prosjekter



Vennligst se i fullskjerm

KAPITTEL I



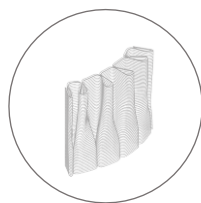
teknologiske
prosjekter

Det første kapittelet fokuserer på integrativ teknologi i arkitektur, i sammenheng med computational design og digital fabrikasjon.

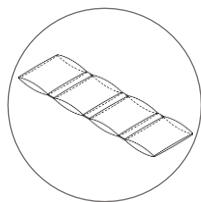
KAPITTEL I



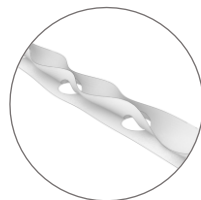
teknologiske
prosjekter



- 1 FUTURE TRADITION
Robotic Clay Printing



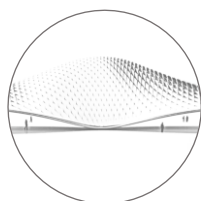
- 2 JOINT FREE MOVEMENT
Fiber Composite Production



- 3 T-BEAM
Bending Active Plates



- 4 FLECTOFOLD
Frame Steel CNC Fabrication



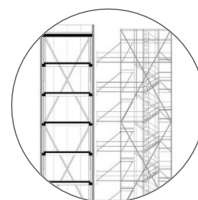
- 5 TRAIN STATION
Flectofold Competition Proposal



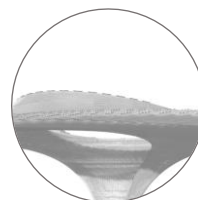
- 6 ITECH RESEARCH DEMONSTRATOR 2019
Biomimetic Adaptive Architecture



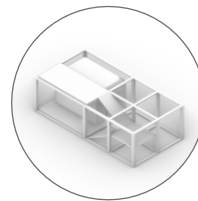
- 7 ELECTROACTIVE SKIN
Electroactive - Textile Production



- 8 ADAPTIVE TOWER
SFB1244 Demonstrator



- 9 MAIN STATION STUTTGART 21
Complex Concrete Reinforcement



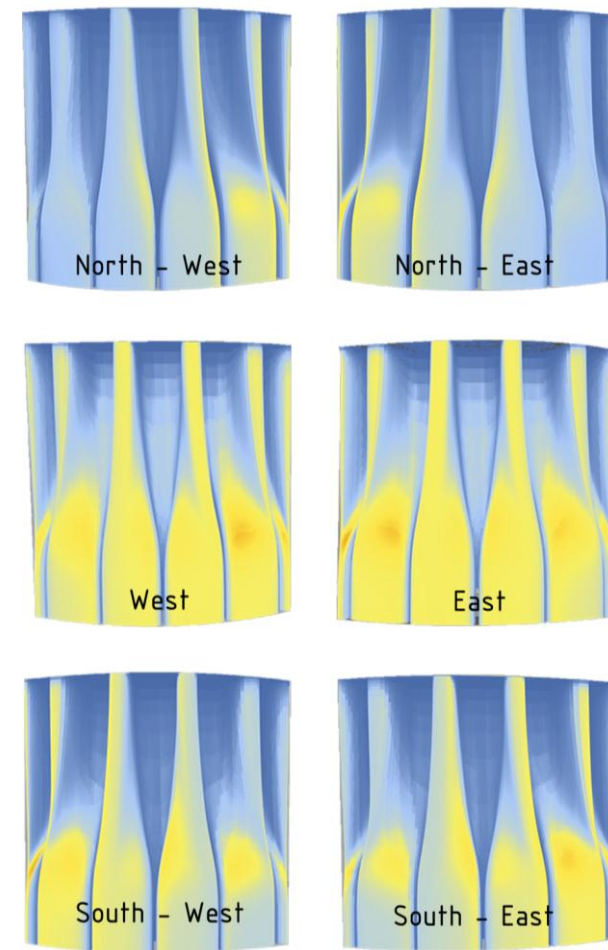
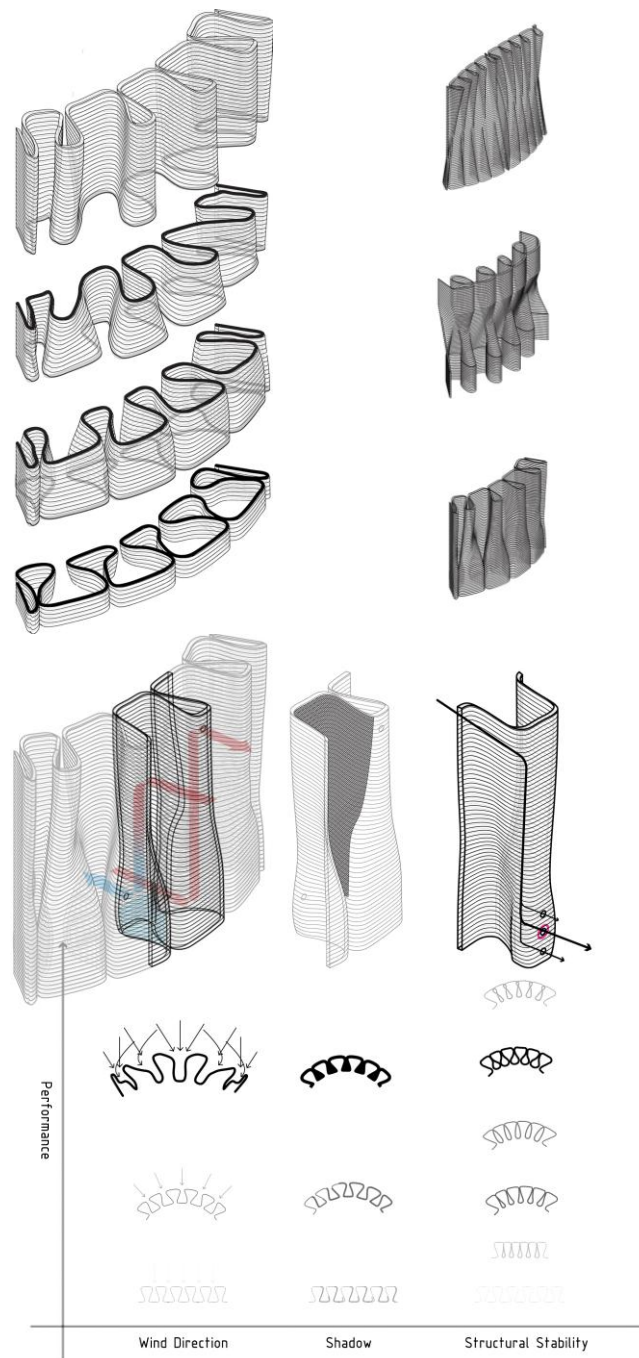
- 10 MODUS HIGH-RISES
Modular Procedural Assembly

work **FUTURE TRADITION**
 Rethink of Vernacular Clay Cooling System

tech KUKA Robotic Clay Printing, Environmental Analysis

role Competition: Future Tradition
 2016 Contemporary Architect Association (CAA), IR

task Entire Process



work **JOINT FREE MOVEMENT**
 Shape-changing Fiber Composite Plates

tech Pneumatic Cushions, Glass Fiber

role ITECH Course: Architectural Biomimetic
 2017 ICD Institute, University of Stuttgart, DE

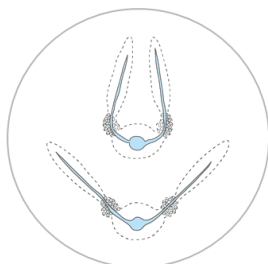
task Entire Process

The aim of this research project is to control the elastic stiffness and bending behavior of a fiber composite plate with pneumatic actuation.

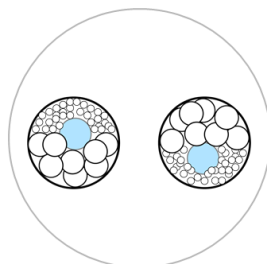
Biological Role Model: Mimosa Pudica



One branch

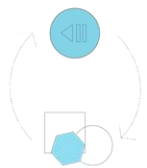


Petiole with leaves



Cells in Pulvini

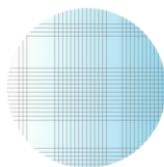
RELEVANT PRINCIPLES FOR ABSTRACTION



Fusion of actuators



Pressure change



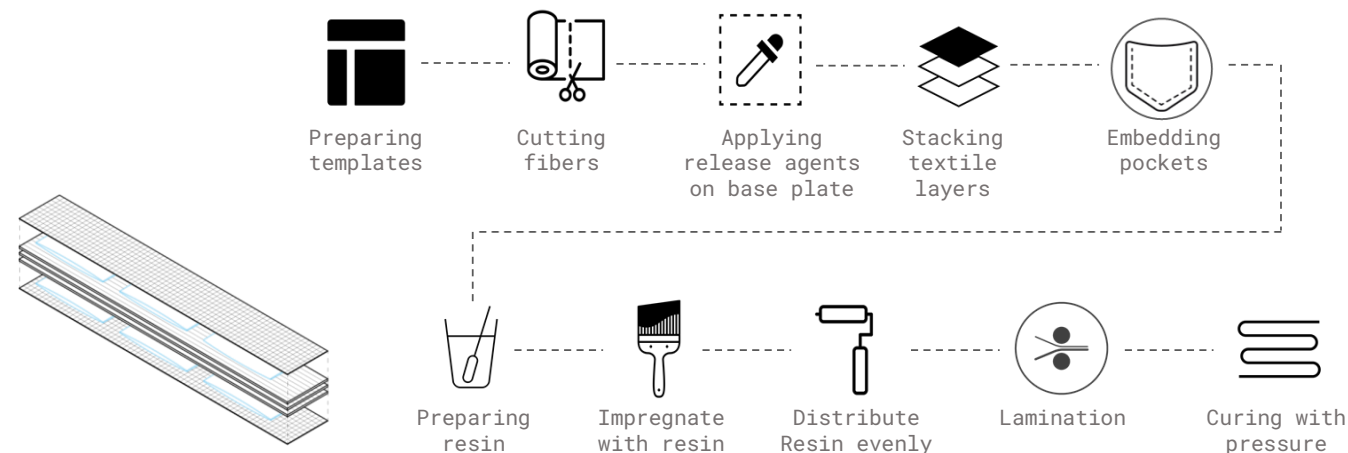
Material gradient



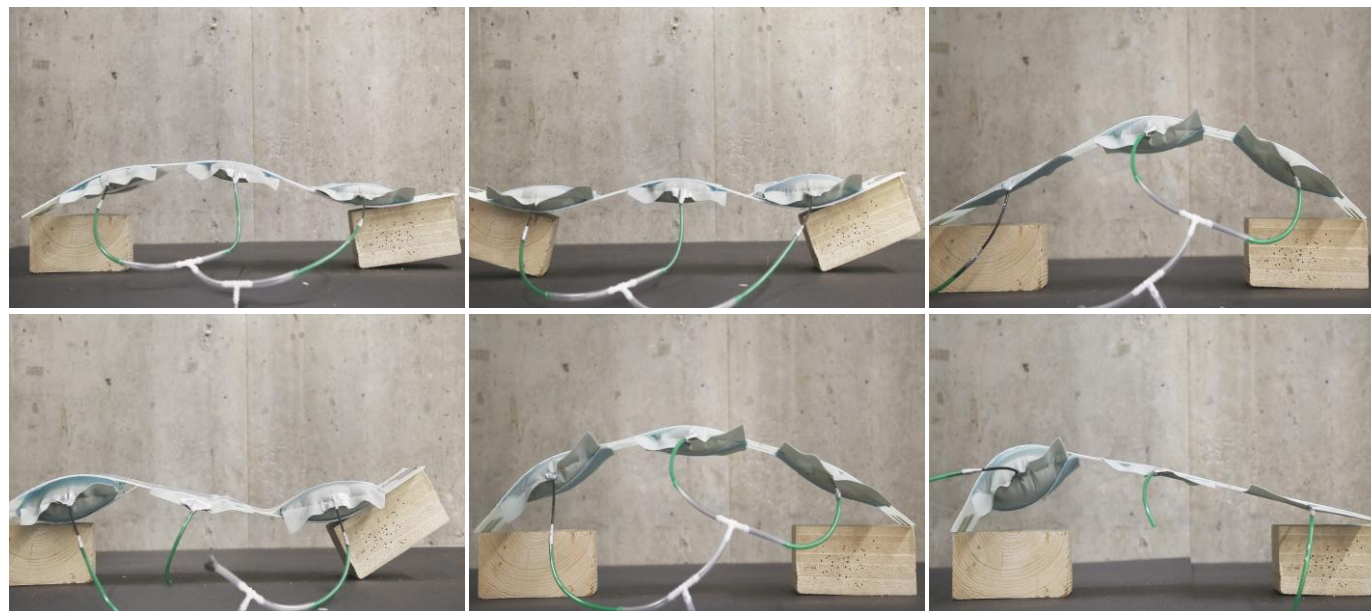
Compliance



Reversibility and freedom



1x Glass fiber woven fabric 0/60
 2x Plastic (pocket)
 3x Glass fiber fabric, unidirectional
 2x Plastic (pocket)
 1x Glass fiber woven fabric 0/60

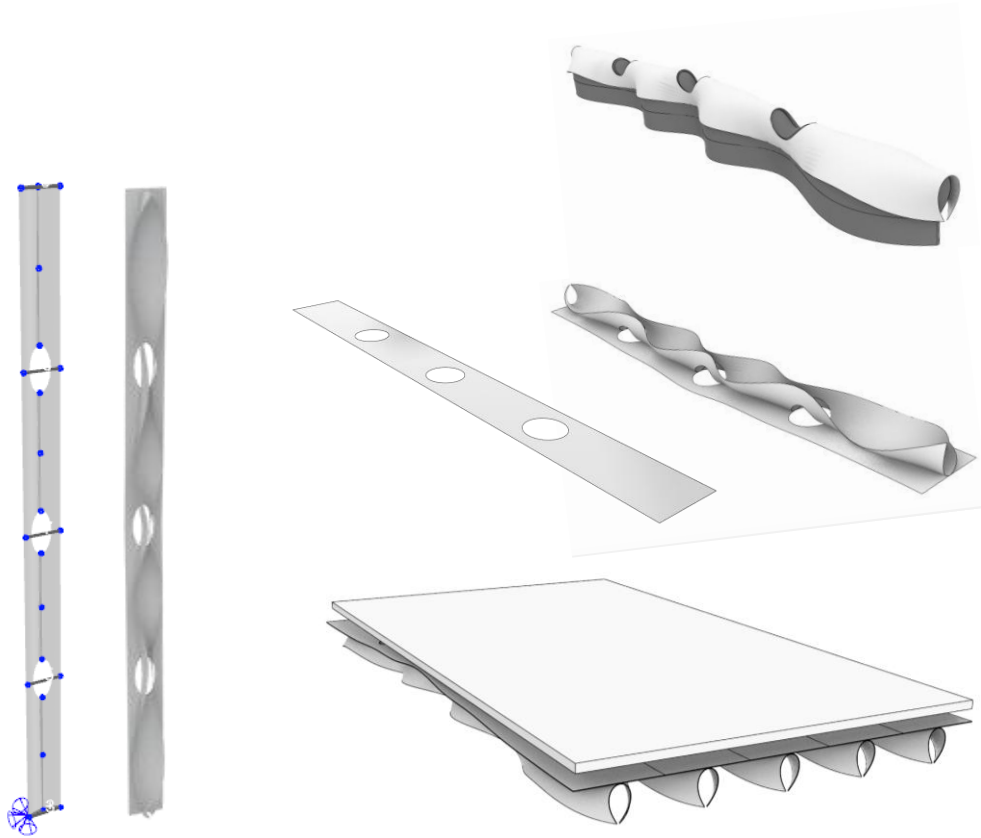


work **T-BEAM**
Curved Bending T Cross-sections

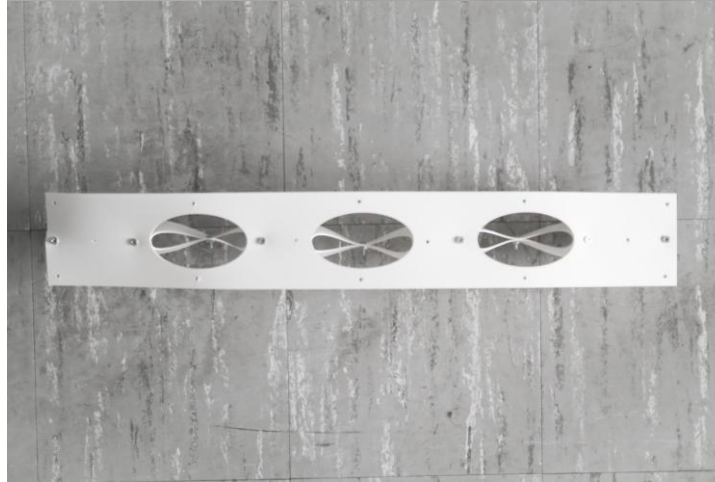
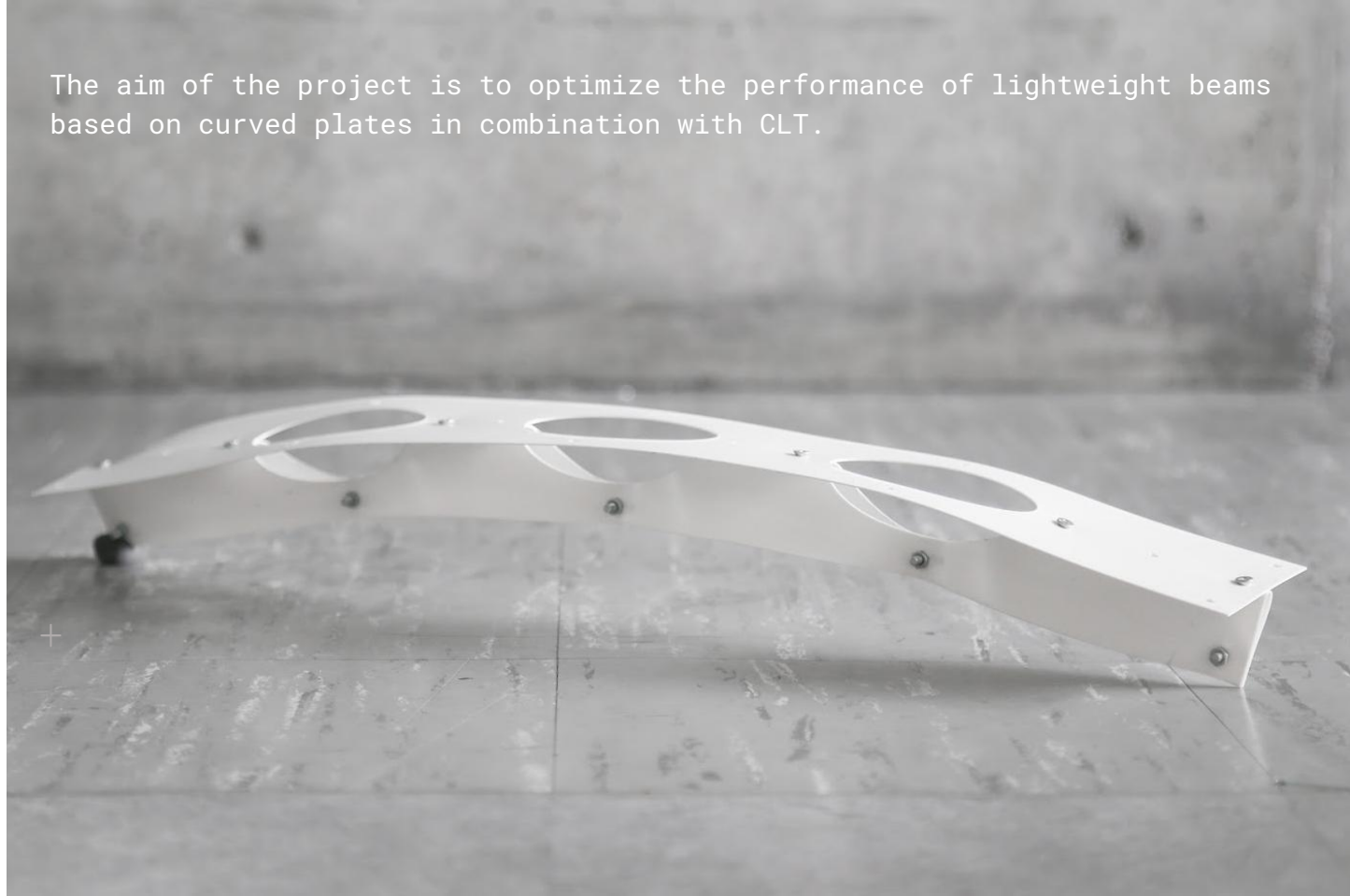
tech Typology Optimization and Manual Prototyping

role ITECH Course: Material and Structure
2018 ITKE Institute, University of Stuttgart, DE

task Entire Process



The aim of the project is to optimize the performance of lightweight beams based on curved plates in combination with CLT.

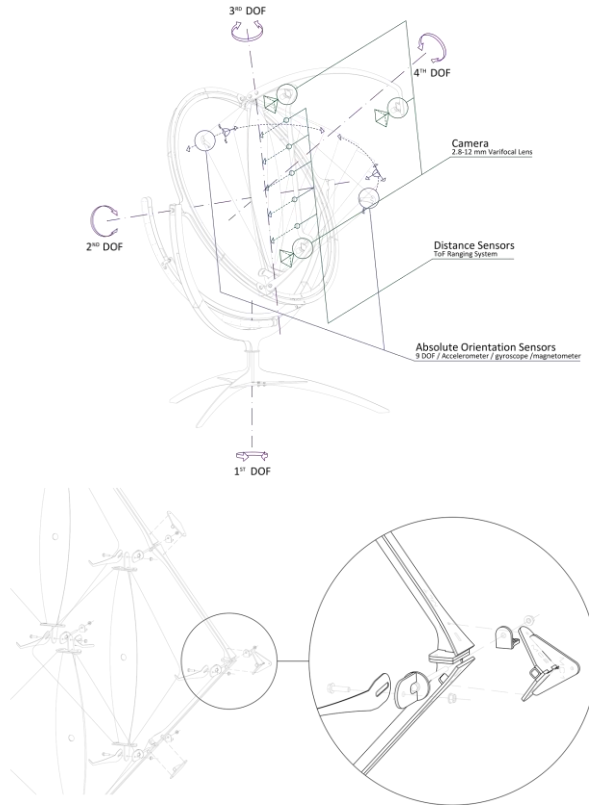


work **FLECTOFOLD**
Demonstrator & Performance Monitoring Device

tech Compliant Mechanism

role Student Assistant (HiWi)
2018 ITKE Institute, University of Stuttgart, DE

task Design Development, Fabrication, Assembly



The aim of this project was to monitor, evaluate and showcase the potential of materially graduated FRP as a material solution for climate adaptive building envelopes. To read more, click [here](#).

work **TRAIN STATION**
FlectoFold Design Development

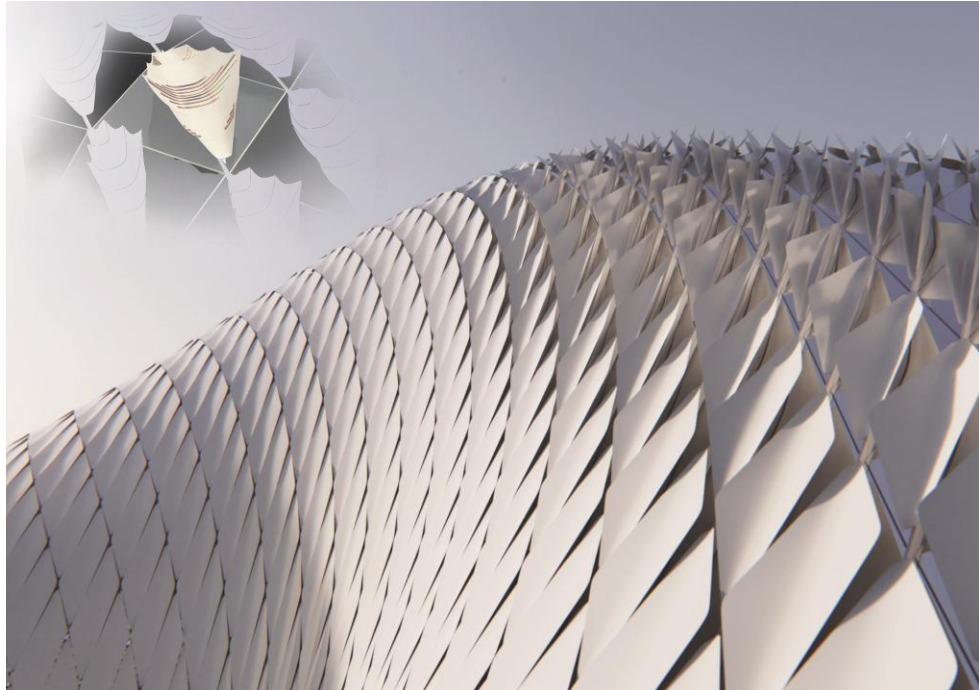
tech Compliant Mechanism

role Competition

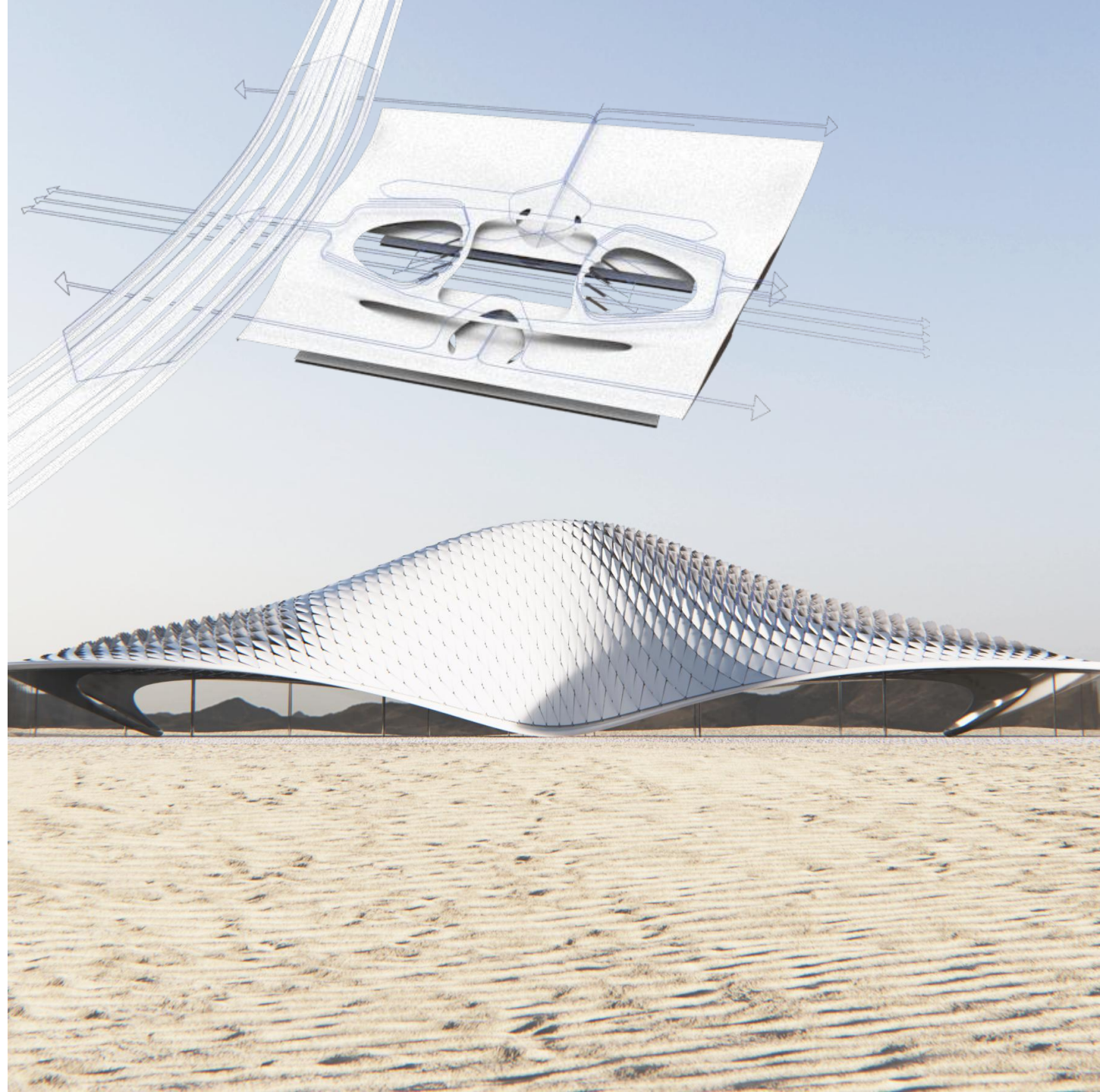
2018 Saman Saffarian Architecture Office, CZ

task Circulation Design and Functioning Diagrams

The aim of this project was to propose an airport using segmented Flectofold shell for environmental adaptation.



Pictures © Saman Saffarian



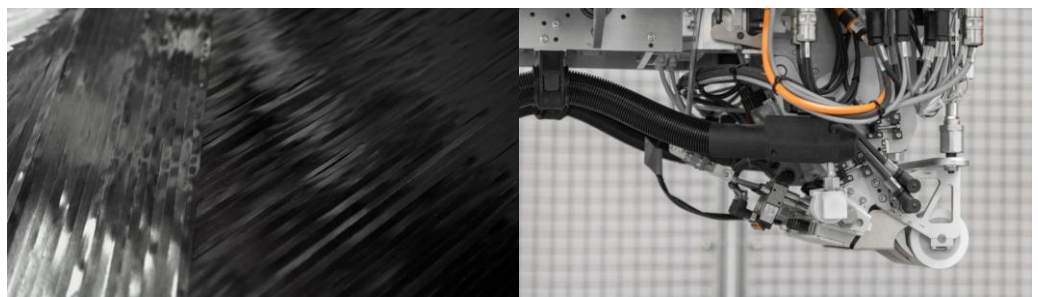
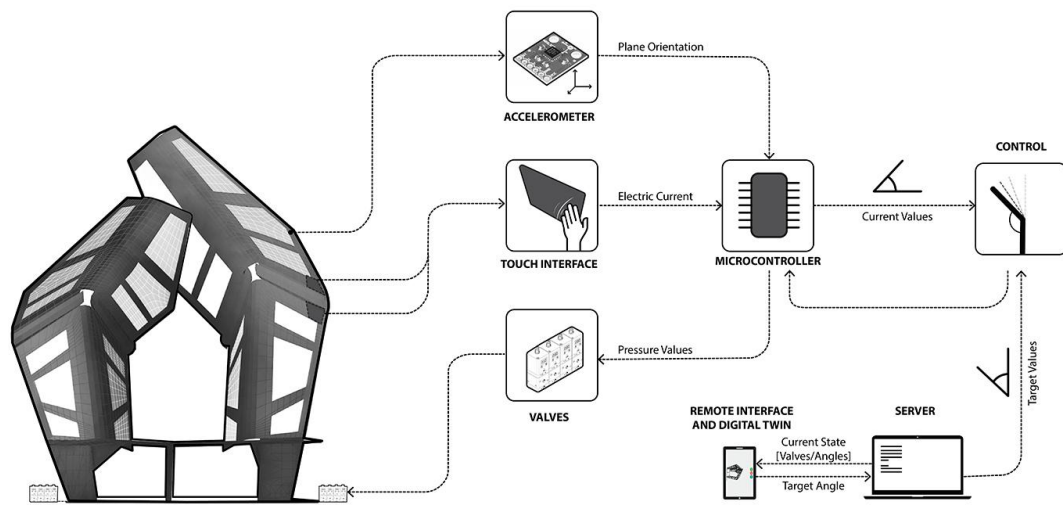
work **ITECH RESEARCH DEMONSTRATOR 2018-19**
Biomimetic Adaptive Architecture

tech Robotic Carbon Fiber Placement, Haptic Adaptive

role ITECH Studio Project (2019)

2019 ICD / ITKE Institutes, University of Stuttgart, DE

task Simulation, Fabrication



Pictures © ITKE / ICD University of Stuttgart



The ITECH research demonstrator 2018/19 investigates large-scale compliant architecture inspired by the folding mechanisms of the Coleoptera Coccinellidae (Ladybug) wings. To read more click [here](#).

work **ELECTROACTIVE SKIN**
Towards Bio-inspired Responsive Envelopes

tech Integration of EAP with Textile

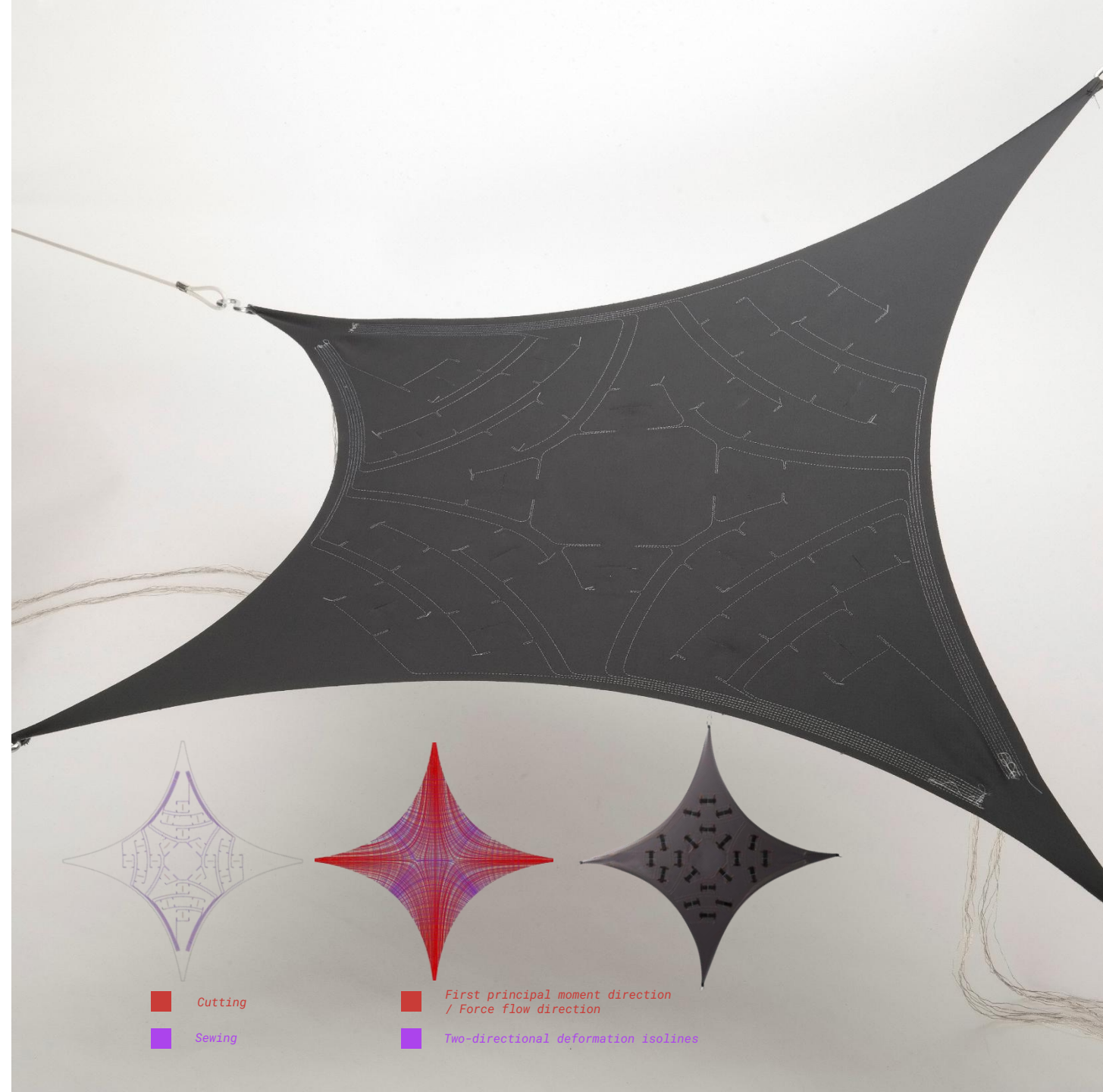
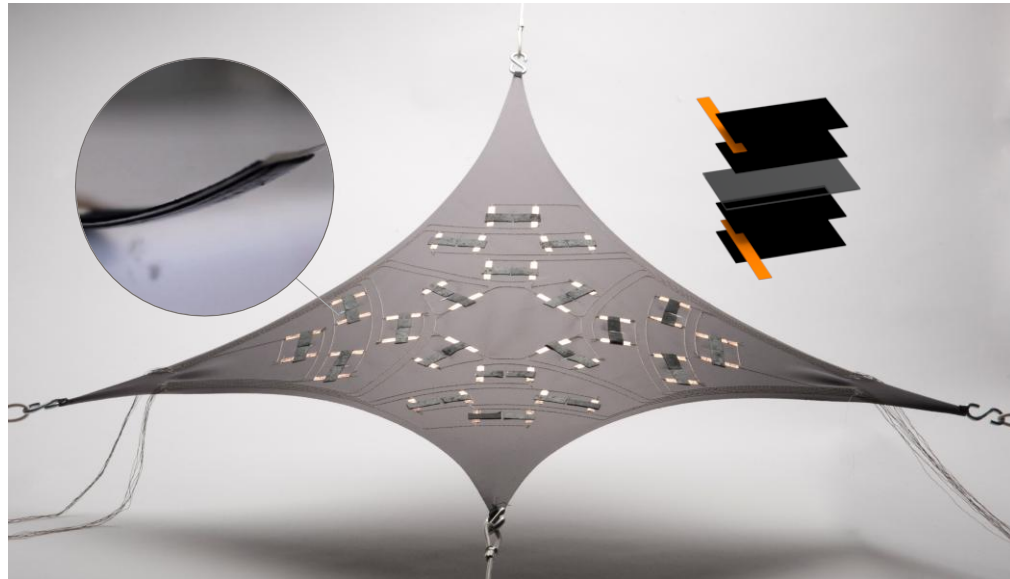
role Master's Thesis (2019)
2019 University of Stuttgart (BioMat/ITKE, ICD), DE
Fraunhofer Institute (IPA), DE
Institute of Aircraft Design (IFB), DE

task Entire Process

The aim of this research is to investigate actuatable apertures in architectural skin through incorporation of ionic electroactive polymer.

To see the video, click [here](#).

To read the published paper, click [here](#).

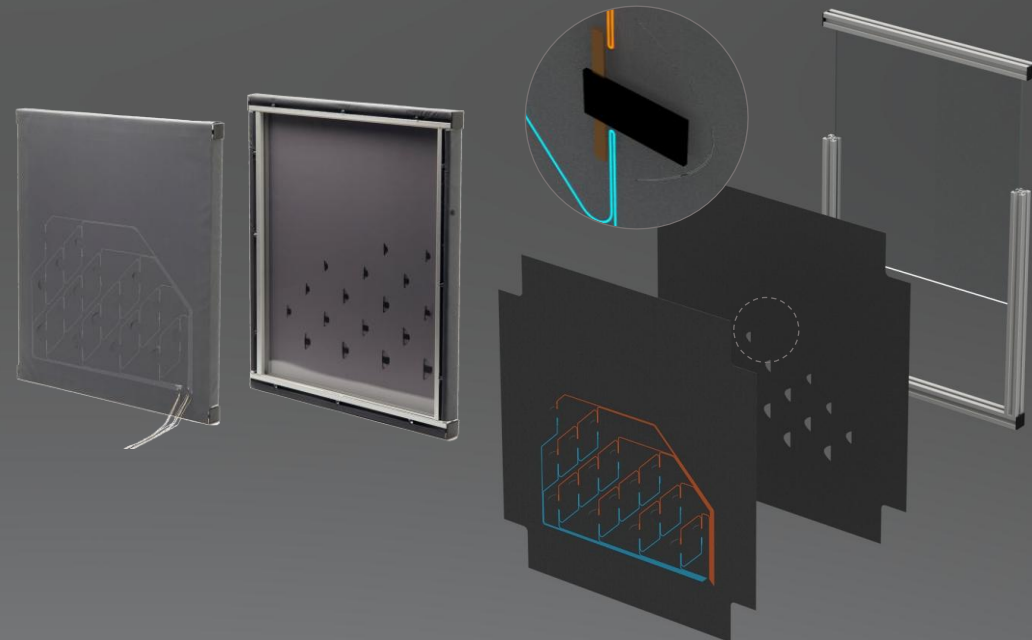
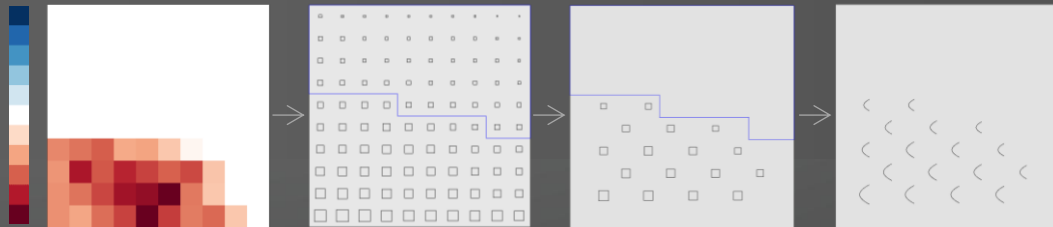


work **ADAPTIVE TOWER**
SFB1244 Adaptive Building Skin Demonstrator

tech Integration of EAP with Textile

role Master's Thesis (2019)
2019 University of Stuttgart (BioMat/ITKE, ICD), DE
Fraunhofer Institute (IPA), DE
Institute of Aircraft Design (IFB), DE

task Entire Process



This research contributes to the ongoing SFB1244 “Adaptive Building Skins and Structures for the Built Environment of Tomorrow”, funded by the German Research Foundation (DFG) under the following schema:

C03 – Electroactive polymer actuators and arrays for switchable breathability in building skins

To read more, click [here](#).



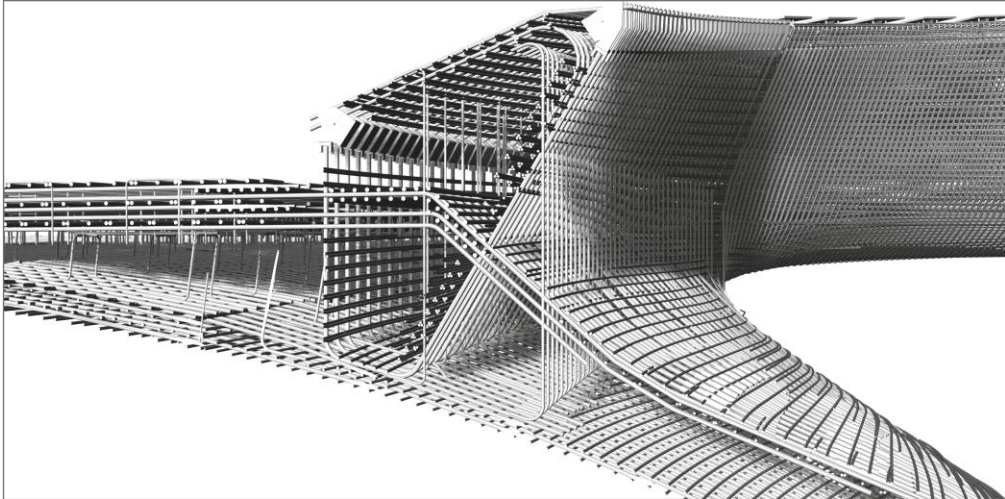
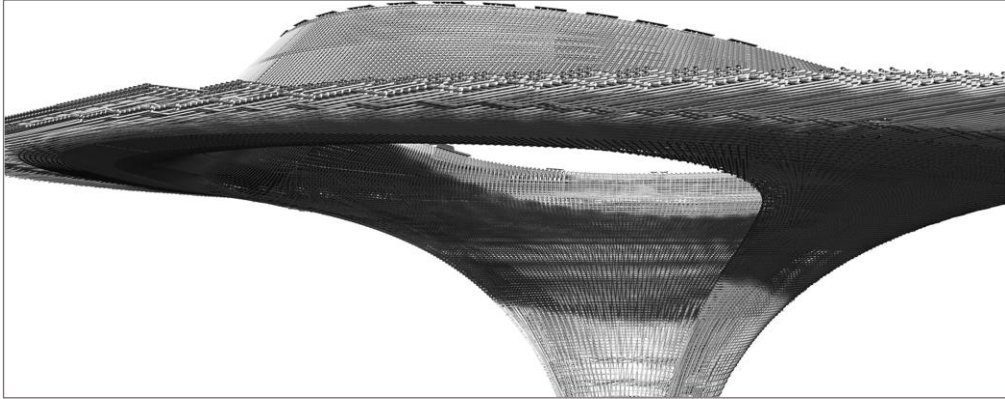
work **MAIN STATION STUTT GART 21**
Reinforced Concrete Shell

tech Computational

role Full-time Employment

2020 *Werner Sobek AG Stuttgart, DE*

task Computational Design



Stuttgart 21 is a transport and urban development project to reorganize the Stuttgart railway junction.



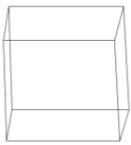
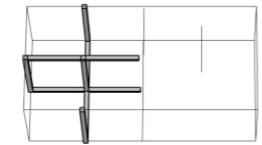
work **MODUS HIGH-RISES**
Modular Procedural Assembly

tech Computational Design

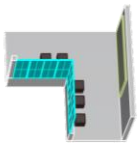
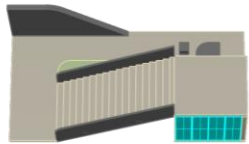
role Individual Project
2020 DesignMorphine Workshop

task Computation

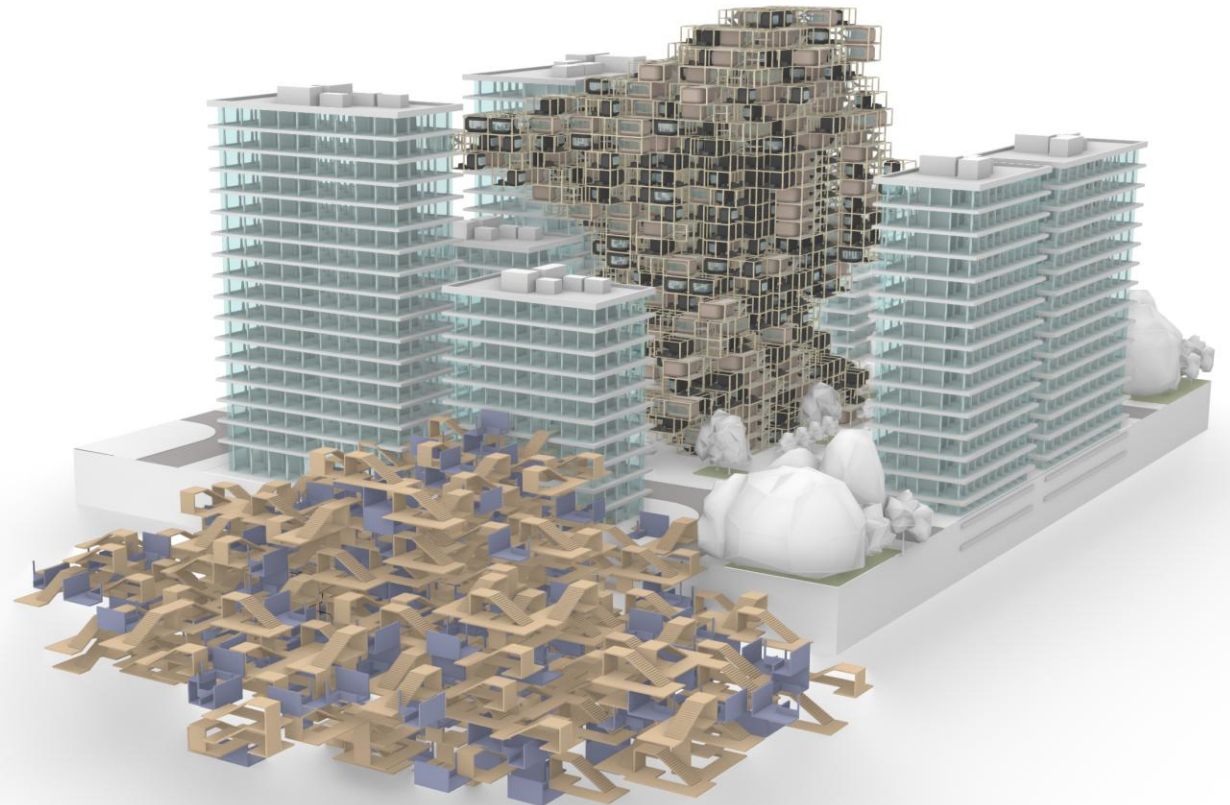
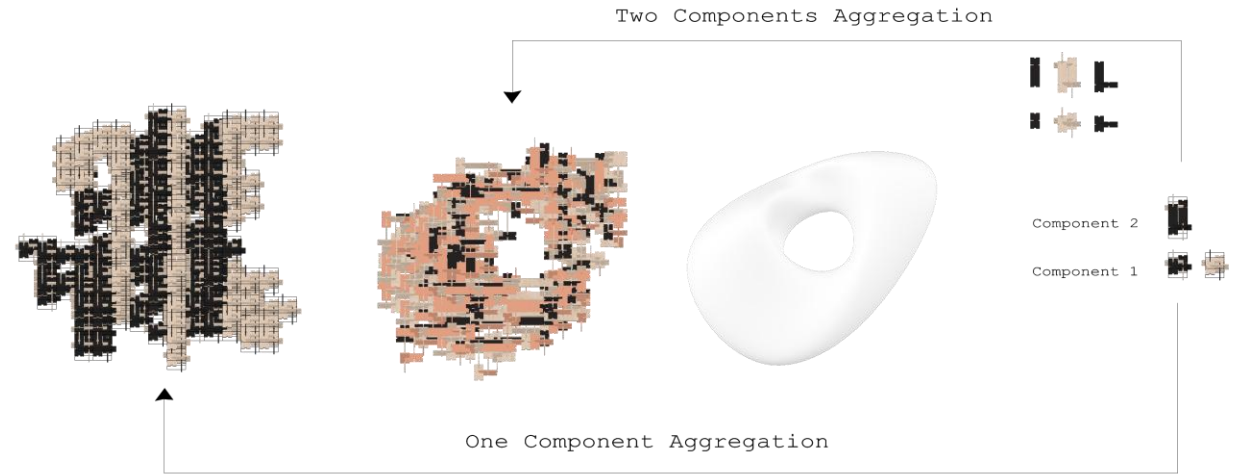
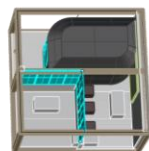
The aim is to create modular procedural modules via WASP within growing transformative spaces, that can react to the changing streams of the information society.



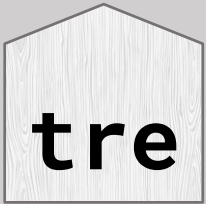
Modules Option 2



Modules Option 1



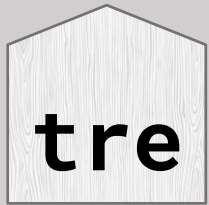
KAPITTEL II



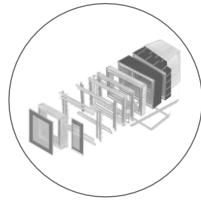
trebaserte
prosjekter

Det andre kapittelet inkluderer **utvalgte innovative treprosjekter** som viser et aspekt ved tre og dets **tilhørende fabrikasjonsteknikker**.

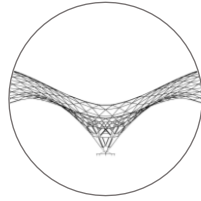
KAPITTEL II



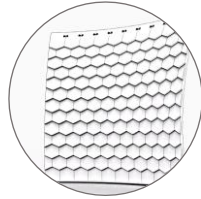
trebaserte
prosjekter



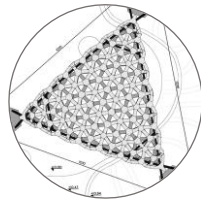
- 1 IBA TIMBER PROTOTYPE HOUSE
Advance CNC Fabrication



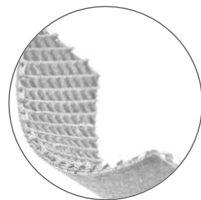
- 2 FUNICULAR TIMBER GRIDSHELL
Form-finding



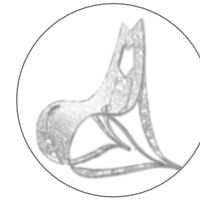
- 3 SHINGLE WALL
KUKA Robotic Fabrication



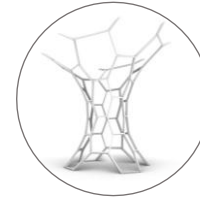
- 4 BIOMAT PAVILION
Vacuum Molding Bio-material



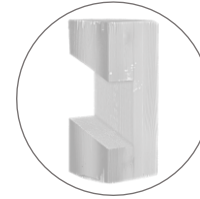
- 5 SMART BIO-STRUCTURE
Wood Filament 4D-Printing



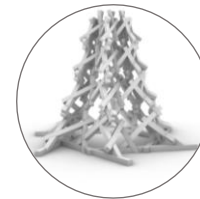
- 6 ODIN CHAIR
Bending Active CNC Plywood



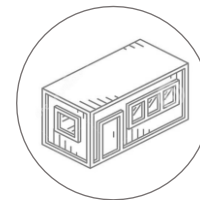
- 7 HEXA TOWER
Triple Scarf Timber Joinery



- 8 INTER-HYGRO-JOINERY
Hygroscopic Analysis via CT Scanning



- 9 INTERLOCKING TOWER
Hundegger Production and AR Assembly



- 10 MODULAR HOUSING
Timber Construction and Manufacturing

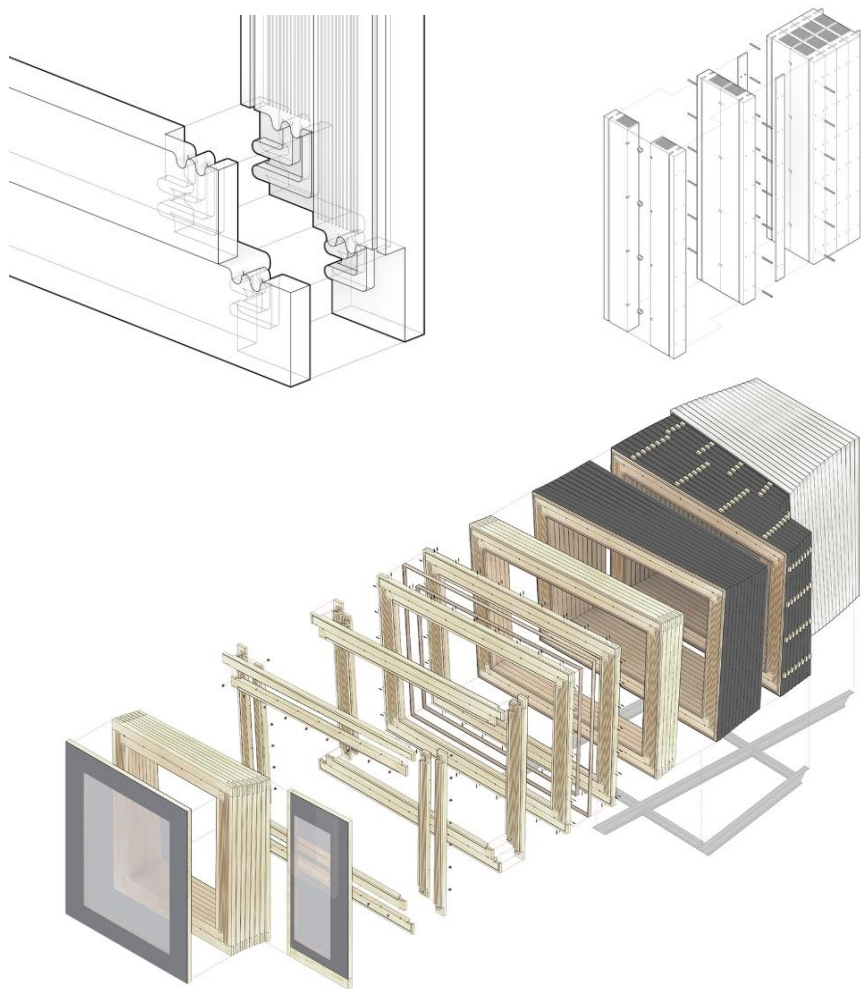
work **IBA TIMBER PROTOTYPE HOUSE**
Digital Log Cabin

tech Computational Design and CNC production

role Student Assistant (HiWi)

2017 ICD Institute, University of Stuttgart, DE

task Construction



Pictures © ICD/ITKE University of Stuttgart

The IBA Timber Prototype House combines the benefits of traditional low-cost timber construction with advances in computational design and fabrication technologies. To read more click [here](#)

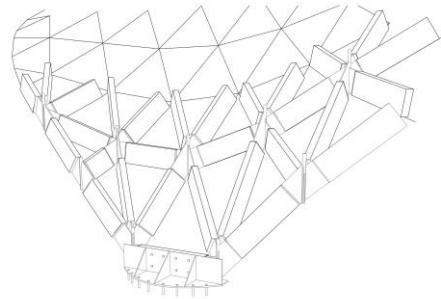
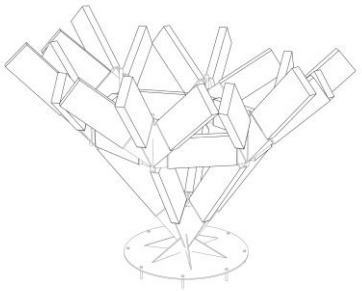
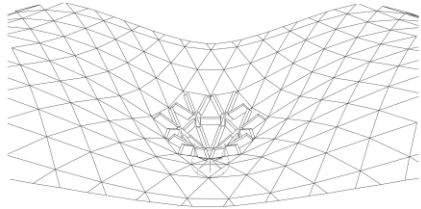
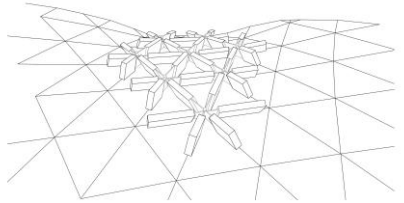
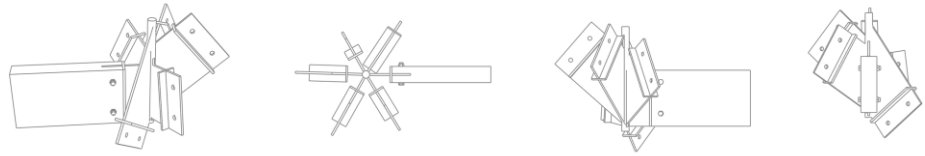


work **FUNICULAR TIMBER GRIDSELL**
Timber Grid-shell with Steel Connectors

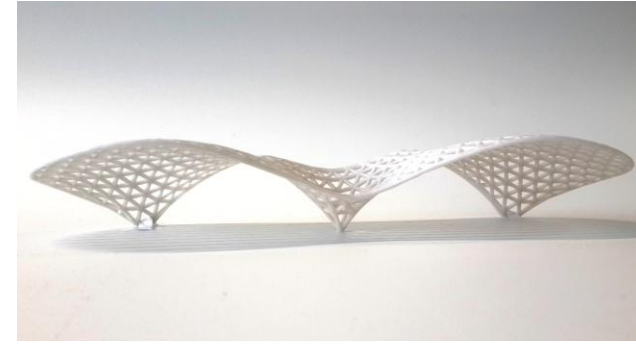
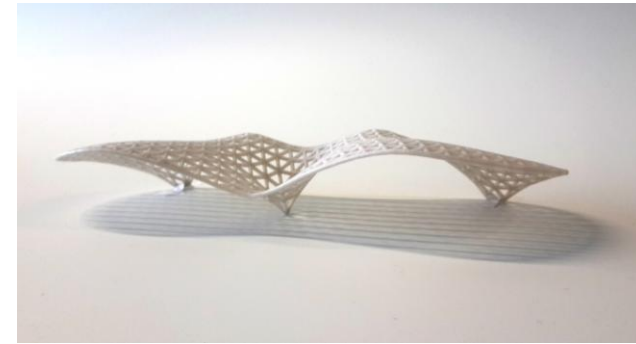
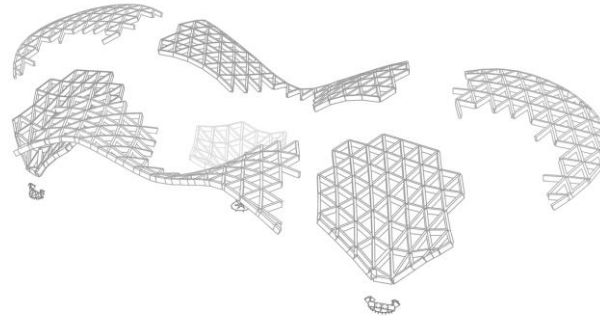
tech Form-finding and Optimization for 3D-printing

role ITECH Course: Form-finding
2017 ITKE Institute, University of Stuttgart, DE

task Entire Process



The aim of this project is to create series of bus stations as a part of VVS in Stuttgart based on funicular timber grid shell.

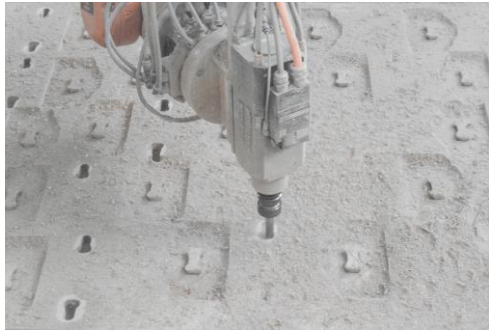
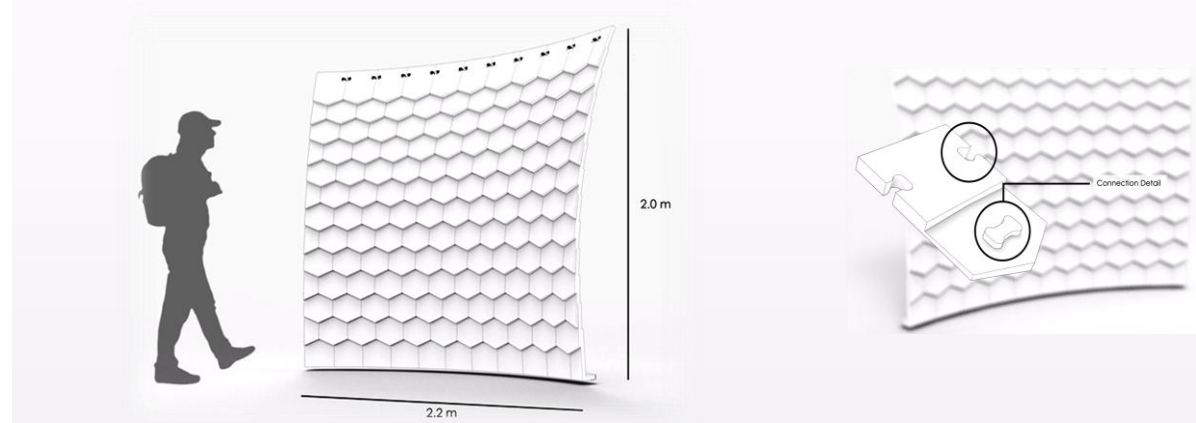


work **SHINGLE WALL**
Shingle Component

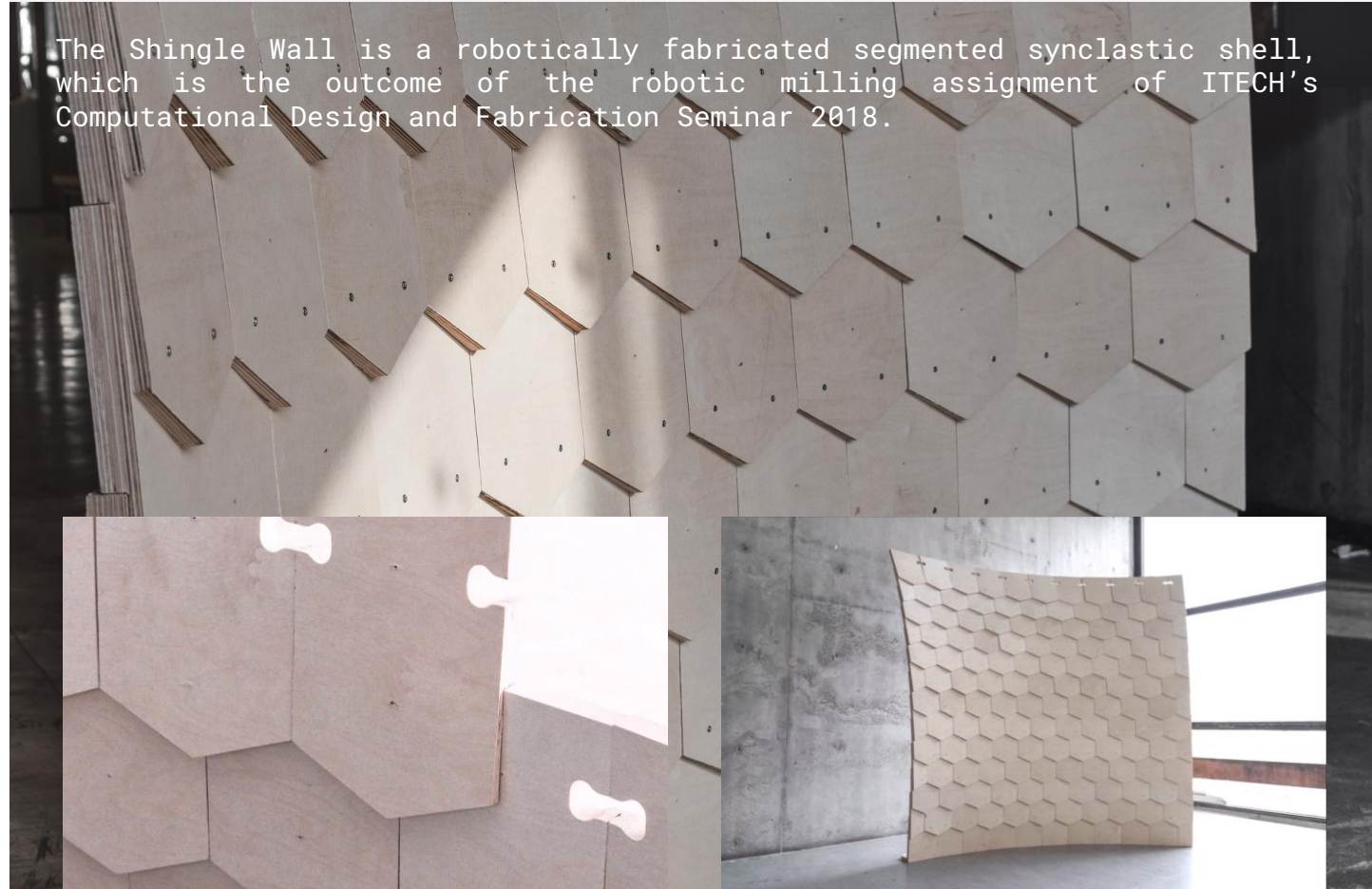
tech Computational Design and Robotic Fabrication

role ITECH Group Project
2018 ICD Institute, University of Stuttgart, DE

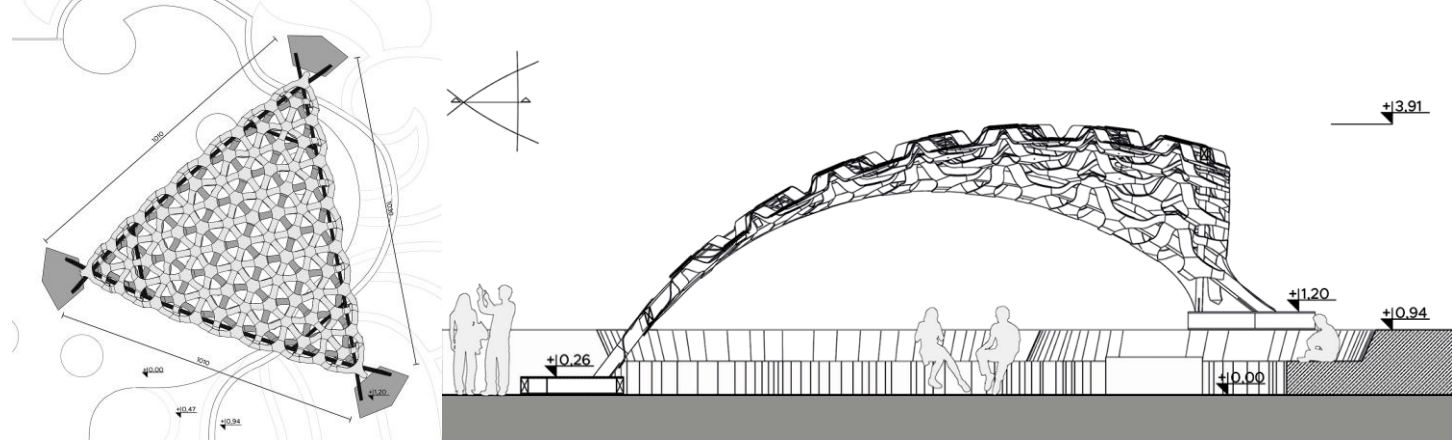
task Entire Process



The Shingle Wall is a robotically fabricated segmented synclastic shell, which is the outcome of the robotic milling assignment of ITECH's Computational Design and Fabrication Seminar 2018.



work **BIOMAT PAVILION**
Bio-composite Flexible Wood Segmental Shell
tech Vacuum Molding Bio-material
role Student Assistant (HiWi)
2018 BioMat Institute, University of Stuttgart, DE
task Manufacturing and Assembly



The pavilion is a double-curved, parametrically designed segmental shell of light, single-curved wood and bio-composite elements supported by three curved crossed wooden beams.



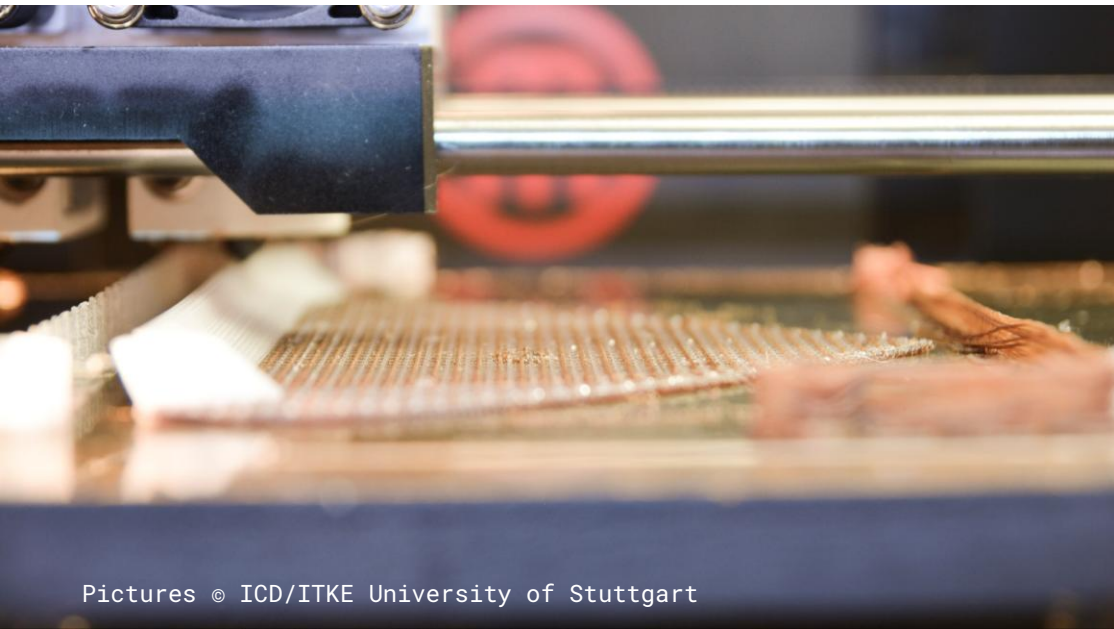
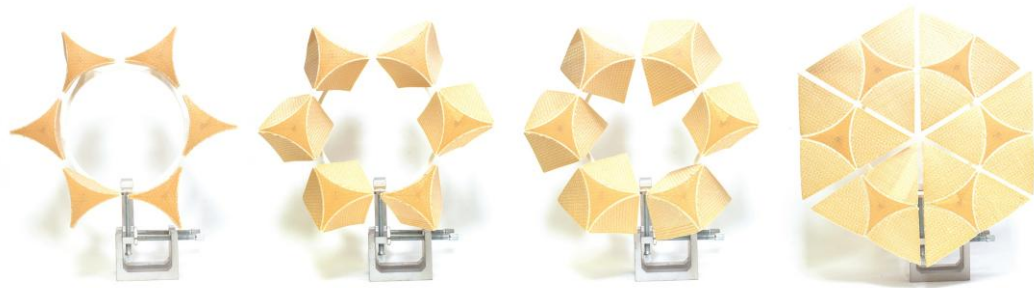
work **SMART BIO STRUCTURE**
Passive Adaptive Façade System

tech 4D Printed Programmable Hygroscopic Wood

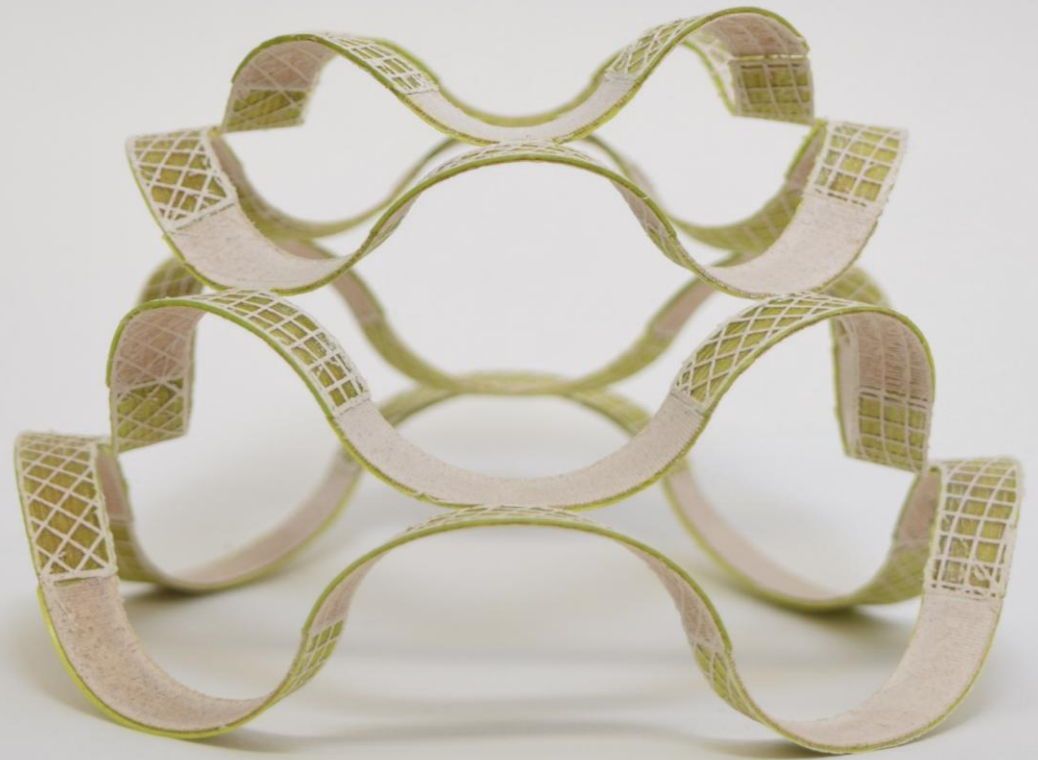
role Student Assistant (HiWi)

2019 ICD Institute, University of Stuttgart, DE

task Simulation and Production



The potentials of programmable 4D-printing of wood filament to create environmentally passive responsive facade systems.

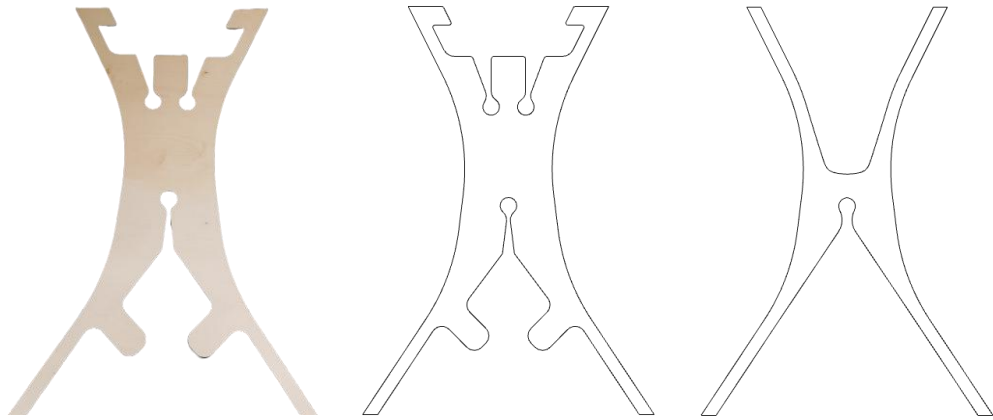


work **ODIN CHAIR**
Bending-active Structure

tech Experimental Prototyping of CNC Plywood

role ITECH Course: Chair Design
2019 BioMat Institute, University of Stuttgart, DE

task Entire Process



The prepared CNC plywood layers were bended and connected in specific points leading to a 3D deformation from 2D sheets which, as a result, provided structural stiffness. To see the video, click [here](#).



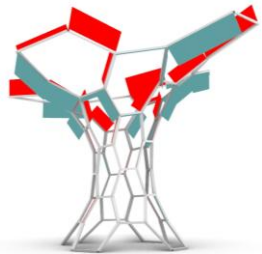
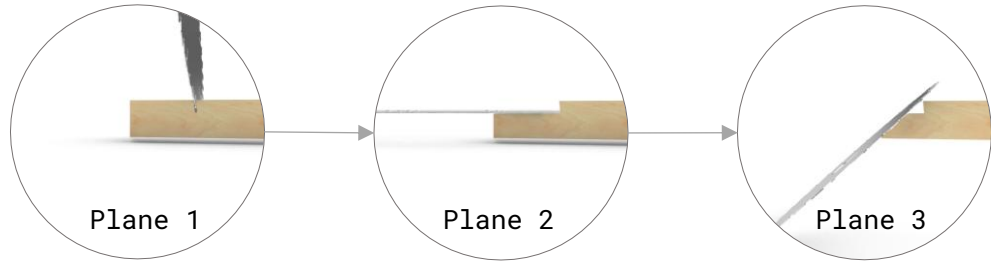
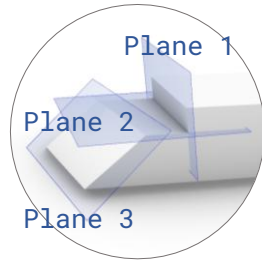
work **HEXA TOWER**
Triple Scarf Timber Joinery

tech FEA and Fabrication Method

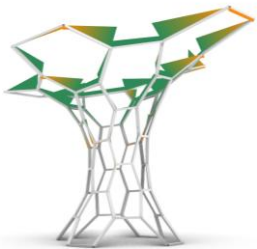
role Research Project

2020 IVB Institute, NTNU i Gjøvik

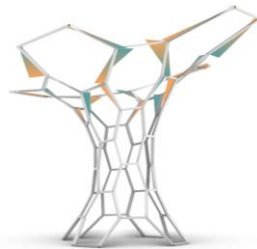
task Entire Process



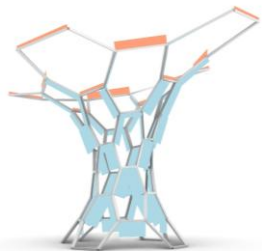
Mx



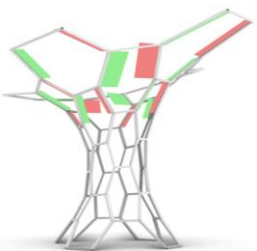
My



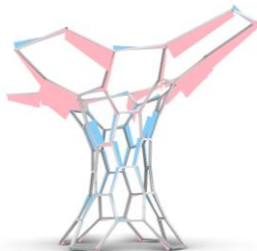
Mz



Nx

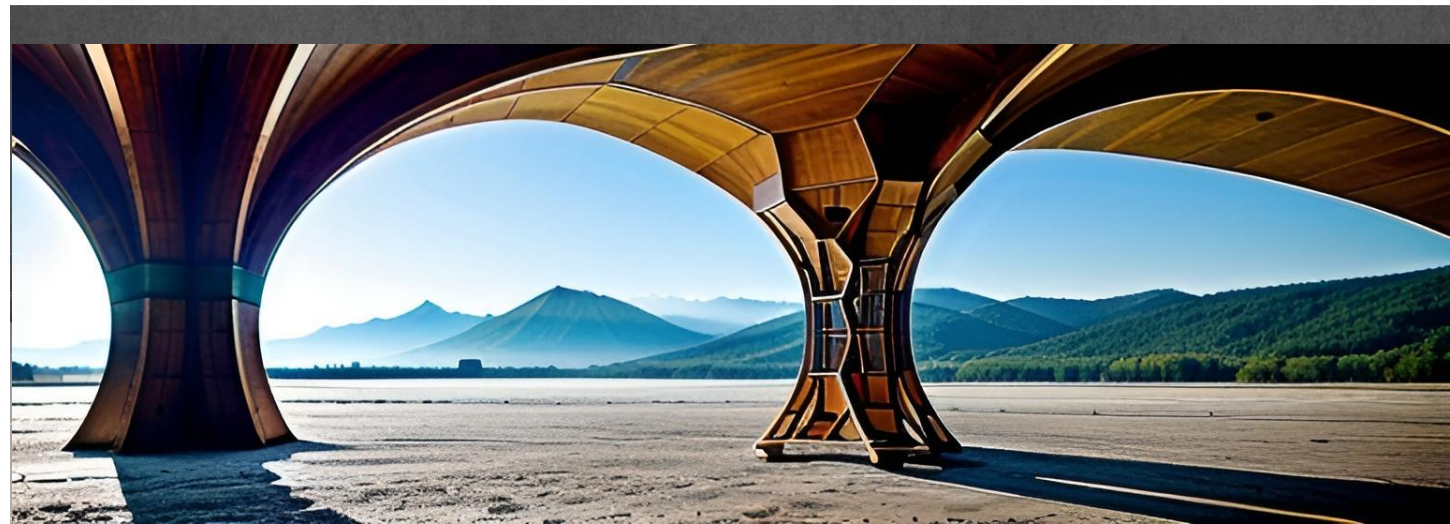
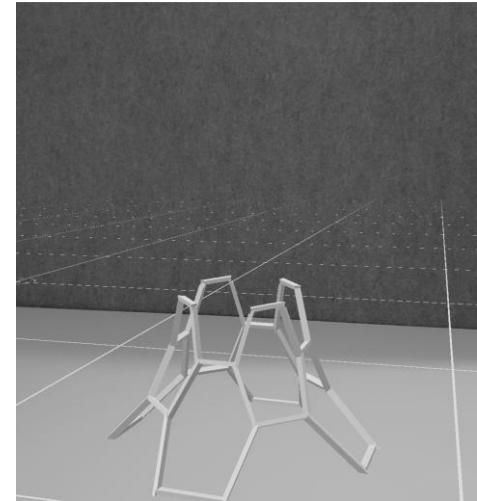
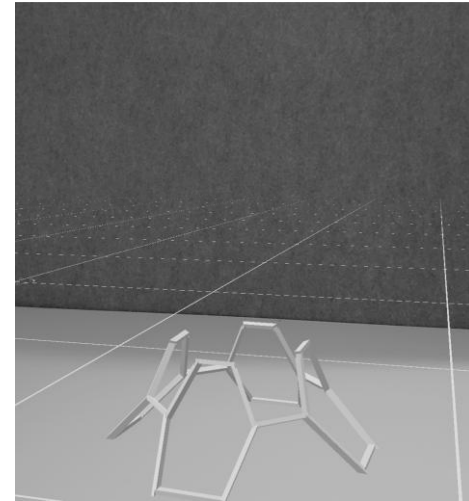
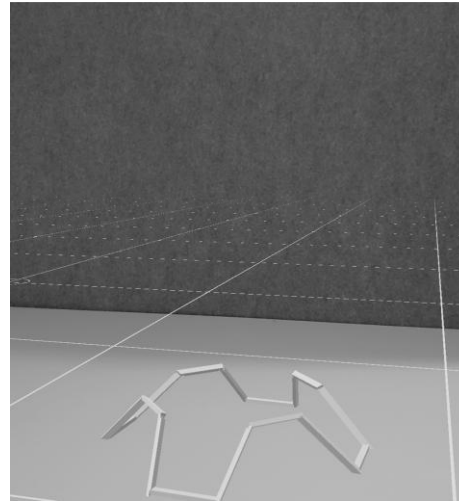
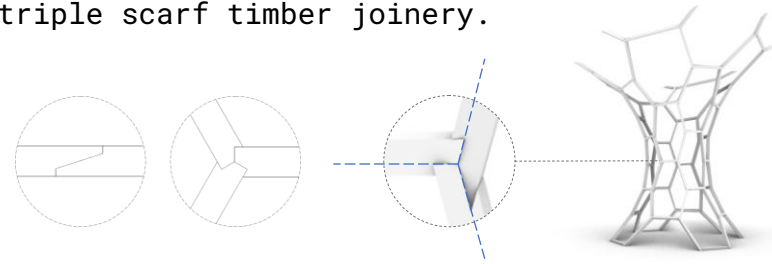


Vy



Vz

The aim of this project is to rethink of vernacular scarf joinery and extend it for triple scarf timber joinery.



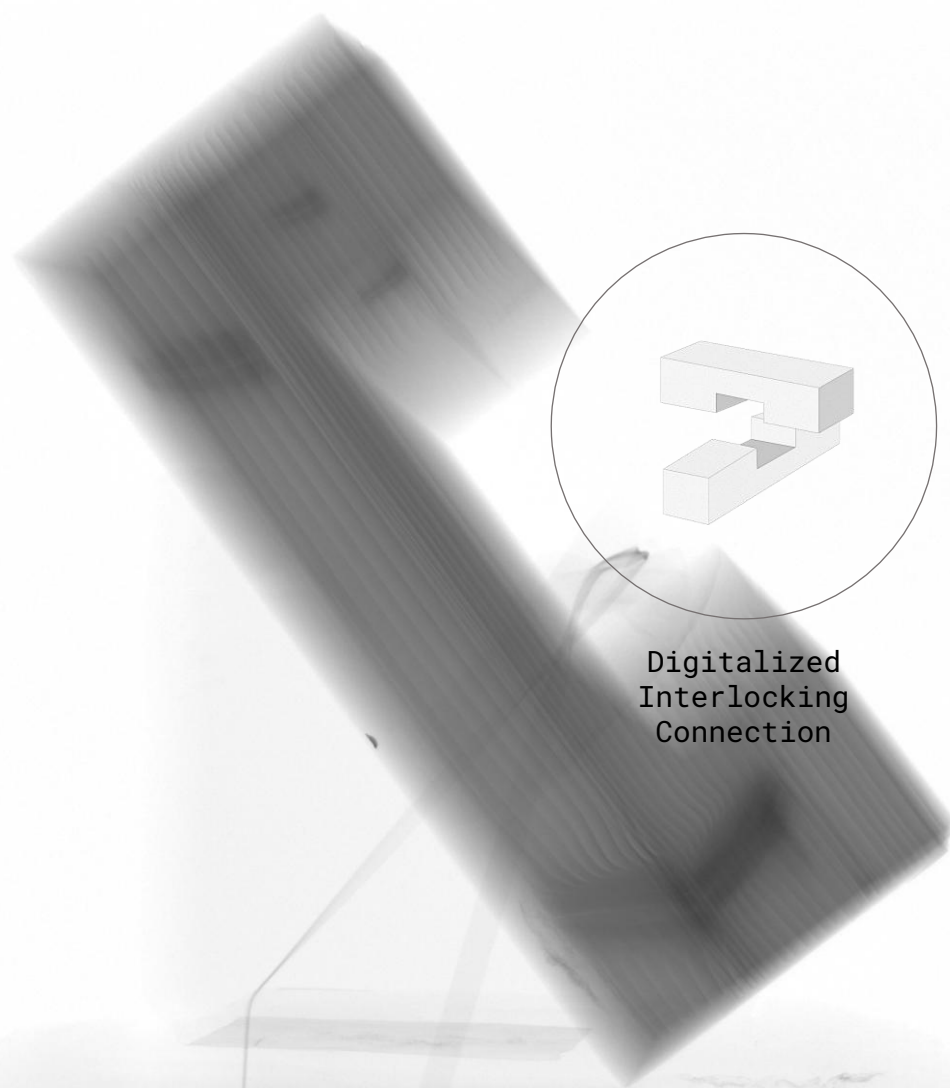
work **INTER-HYGRO-JOINERY**
Encapsulating Interlocking Timber Joinery

tech Hygroscopic Study of Wooden Joinery via CT Scanning

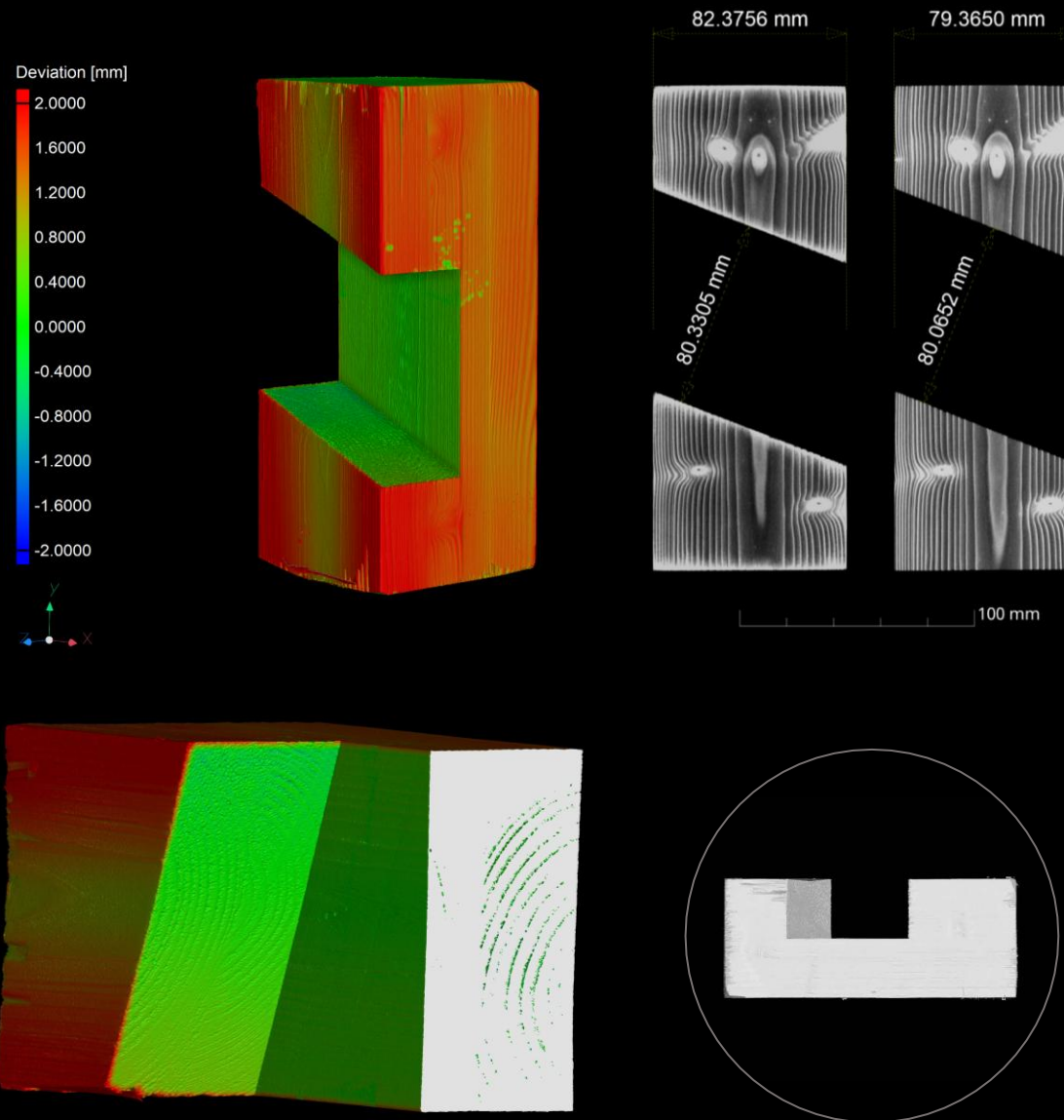
role Research Project

2021 IVB Institute, NTNU i Gjøvik, NO

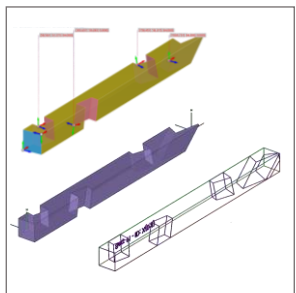
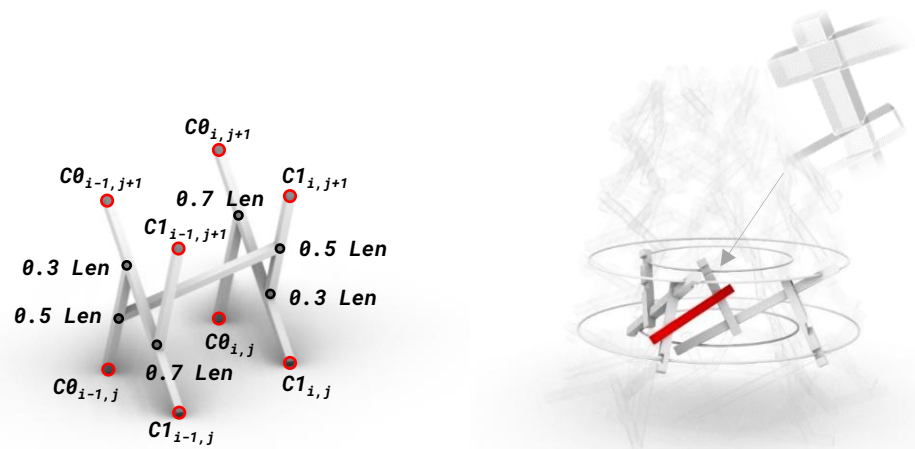
task Entire Process



Investigation of wood's hygroscopicity via CT scanner to reinforce the structural capacity of interlocking connections.



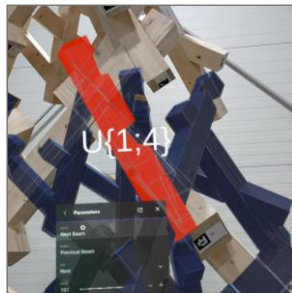
work **INTERLOCKING TOWER**
Sequential Circular Augmented Construction
tech Hundegger Production and Sequential AR Assembly
role Research Project
2022 IVB Institute, NTNU i Gjøvik & Østlaft AG, NO
task Entire Process



Computational
Design



Hundegger
Fabrication



Augmented
Assembly

How innovative structures are realized through synthesizing of interlocking joinery, and how augmented-reality assembly could enable the circular economy in wood construction (?)



work **MODULAR HOUSING**
Serial Timber Housing

tech Modular Timber Construction and Production

role Full-time Employment

2023 *timpla GmbH, Eberswalde-Berlin, DE (timpla.eu)*

task Timber Construction Design

Germany's largest wooden module factory ...
We are industrializing wooden modular construction to establish it as the strongest and most sustainable alternative for residential and commercial construction.



Foto: Gataric Fotografie



Foto: Ruedi Walti



Foto: Stefan Hofmann, Biel